

## SECR2043 OPERATING SYSTEMS

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 Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

### ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write **"No output"** instead.

|    |                                                                                   |                                                                                                                                                                                                                                                                                               |
|----|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Display your current working directory.                                           |                                                                                                                                                                                                                                                                                               |
|    | Command                                                                           | Output                                                                                                                                                                                                                                                                                        |
|    | pwd                                                                               | <pre>rick@secr2043:~\$ pwd /home/rick rick@secr2043:~\$</pre>                                                                                                                                                                                                                                 |
| 2. | List all the files and directories in your home directory, including hidden ones. |                                                                                                                                                                                                                                                                                               |
|    | Command                                                                           | Output                                                                                                                                                                                                                                                                                        |
|    | ls -a ~                                                                           | <pre>rick@secr2043:~\$ ls -a ~ .      .bashrc      .sudo_as_admin_successful  practice.txt ..     .cache       file1.txt                  sample.txt .bash_history .local      hello.txt                  sample.txt.save .bash_logout .profile    nano.1640.save             workspace</pre> |
| 3. | Create a new directory named "os_lab" in your home directory.                     |                                                                                                                                                                                                                                                                                               |
|    | Command                                                                           | Output                                                                                                                                                                                                                                                                                        |
|    | mkdir os_lab                                                                      | <pre>rick@secr2043:~\$ ls file1.txt  nano.1640.save  sample.txt      workspace hello.txt  practice.txt   sample.txt.save rick@secr2043:~\$ mkdir os_lab rick@secr2043:~\$ ls file1.txt  nano.1640.save  practice.txt  sample.txt.save hello.txt  os_lab         sample.txt    workspace</pre> |
| 4. | Change your current working directory to "os_lab".                                |                                                                                                                                                                                                                                                                                               |
|    | Command                                                                           | Output                                                                                                                                                                                                                                                                                        |
|    | cd os_lab                                                                         | <pre>rick@secr2043:~\$ cd os_lab rick@secr2043:~/os_lab\$ pwd /home/rick/os_lab rick@secr2043:~/os_lab\$  </pre>                                                                                                                                                                              |

|     |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.  | Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|     | touch hello.txt<br>echo "Hello World!" >> hello.txt                                             | rick@secr2043:~/os_lab\$ touch hello.txt<br>rick@secr2043:~/os_lab\$ echo "Hello World!" >> hello.txt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6.  | Display the content of "hello.txt".                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|     | cat hello.txt                                                                                   | rick@secr2043:~/os_lab\$ cat hello.txt<br>Hello World!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7.  | Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|     | cp hello.txt hello_copy.txt                                                                     | rick@secr2043:~/os_lab\$ cp hello.txt hello_copy.txt<br>rick@secr2043:~/os_lab\$ ls<br>hello.txt hello_copy.txt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 8.  | Rename "hello_copy.txt" to "hello_bak.txt".                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|     | mv hello_copy.txt hello_bak.txt                                                                 | rick@secr2043:~/os_lab\$ mv hello_copy.txt hello_bak.txt<br>rick@secr2043:~/os_lab\$ ls<br>hello.txt hello_bak.txt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9.  | Delete "hello.txt".                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|     | rm hello.txt                                                                                    | rick@secr2043:~/os_lab\$ rm hello.txt<br>rick@secr2043:~/os_lab\$ ls<br>hello_bak.txt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 10. | Display your home directory with additional information, together with subdirectories contents. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|     | ls -R /home                                                                                     | rick@secr2043:~/os_lab\$ ls -R /home<br>/home:<br>farkhana mahyuddin rick student<br>ls: cannot open directory '/home/farkhana': Permission denied<br>ls: cannot open directory '/home/mahyuddin': Permission denied<br><br>/home/rick:<br>file1.txt nano.1640.save practice.txt sample.txt.save<br>hello.txt os_lab sample.txt workspace<br><br>/home/rick/os_lab:<br>hello_bak.txt<br><br>/home/rick/workspace:<br>argument hello_world.c kernel_info keyboard.c<br>argument.c hello_world.cpp kernel_info.c multiply<br>hello_world input_keyboard.c keyboard multiply.c<br>ls: cannot open directory '/home/student': Permission denied<br>rick@secr2043:~/os_lab\$ |
|     | <b>Total Mark:</b>                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

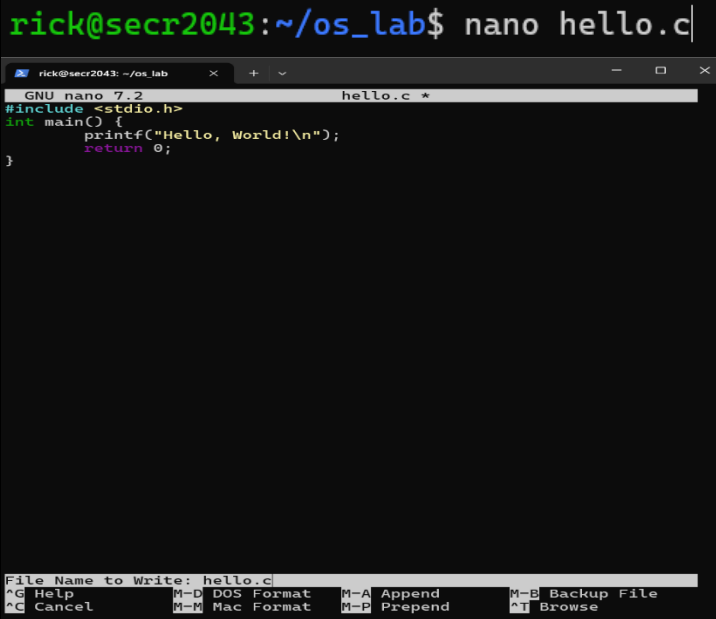
For each question, write down the command and the output (if any) in the answer sheet.

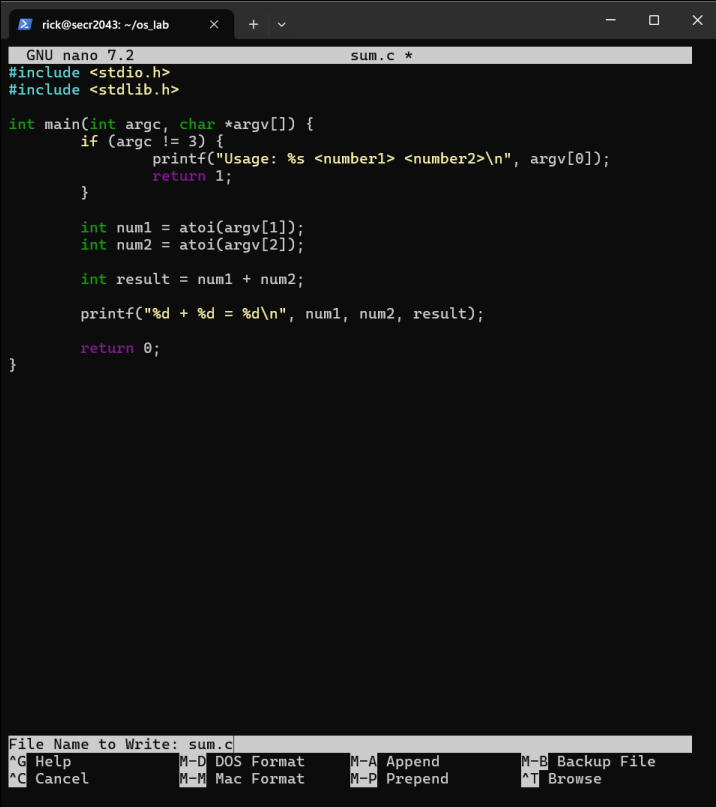
|    |                                                                                                                    |                                                                                                                                                                     |
|----|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Create an empty file named "hello.txt" in the current working directory.                                           |                                                                                                                                                                     |
|    | Command                                                                                                            | Output                                                                                                                                                              |
|    | touch hello.txt                                                                                                    | <pre>rick@secr2043:~/os_lab\$ touch hello.txt rick@secr2043:~/os_lab\$ ls hello.txt  hello_bak.txt</pre>                                                            |
| 2. | Write the text "Hello World" to "hello.txt" using echo.                                                            |                                                                                                                                                                     |
|    | Command                                                                                                            | Output                                                                                                                                                              |
|    | echo "Hello World" > hello.txt                                                                                     | <pre>rick@secr2043:~/os_lab\$ echo "Hello World" &gt; hello.txt</pre>                                                                                               |
| 3. | Append the text "This is a test" to "hello.txt" using echo.                                                        |                                                                                                                                                                     |
|    | Command                                                                                                            | Output                                                                                                                                                              |
|    | echo "This is a test" >> hello.txt                                                                                 | <pre>rick@secr2043:~/os_lab\$ echo "This is a test" &gt;&gt; hello.txt</pre>                                                                                        |
| 4. | Display the contents of "hello.txt" using cat.                                                                     |                                                                                                                                                                     |
|    | Command                                                                                                            | Output                                                                                                                                                              |
|    | cat hello.txt                                                                                                      | <pre>rick@secr2043:~/os_lab\$ cat hello.txt Hello World This is a test</pre>                                                                                        |
| 5. | Create another file named "bye.txt" with the text "Goodbye World" using echo.                                      |                                                                                                                                                                     |
|    | Command                                                                                                            | Output                                                                                                                                                              |
|    | touch bye.txt<br>echo "Goodbye World" >> bye.txt                                                                   | <pre>rick@secr2043:~/os_lab\$ touch bye.txt rick@secr2043:~/os_lab\$ echo "Goodbye World" &gt;&gt; bye.txt rick@secr2043:~/os_lab\$ cat bye.txt Goodbye World</pre> |
| 6. | Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.                                  |                                                                                                                                                                     |
|    | Command                                                                                                            | Output                                                                                                                                                              |
|    | cat hello.txt bye.txt                                                                                              | <pre>rick@secr2043:~/os_lab\$ cat hello.txt bye.txt Hello World This is a test Goodbye World rick@secr2043:~/os_lab\$  </pre>                                       |
| 7. | Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat. |                                                                                                                                                                     |
|    | Command                                                                                                            | Output                                                                                                                                                              |
|    | cat hello.txt bye.txt > greeting.txt                                                                               | <pre>rick@secr2043:~/os_lab\$ cat hello.txt bye.txt &gt; greeting.txt</pre>                                                                                         |

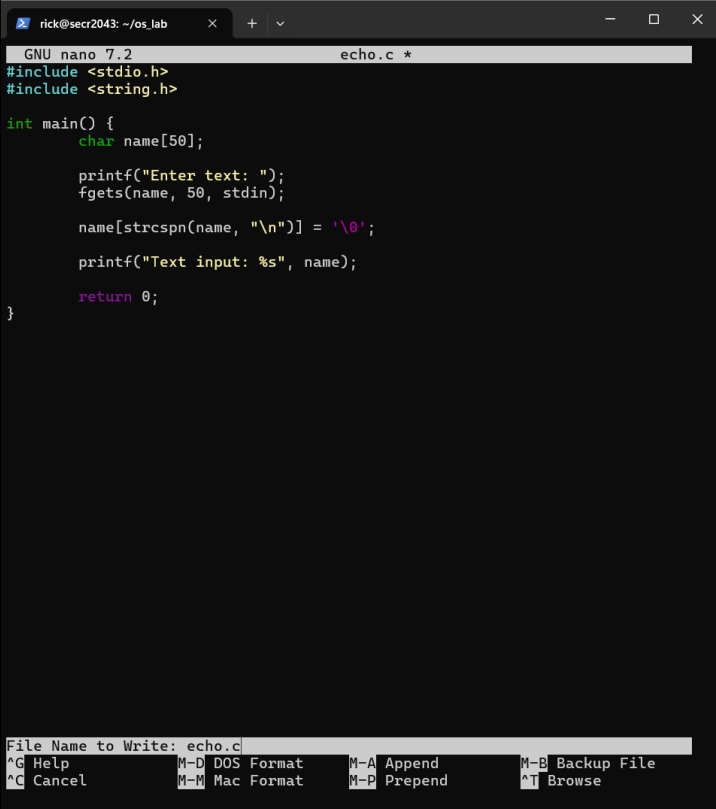
|     |                                                                                                 |                                                                                                                                                                       |
|-----|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.  | Display the contents of "greeting.txt" using cat.                                               |                                                                                                                                                                       |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                         |
|     | cat greeting.txt                                                                                | <pre>rick@secr2043:~/os_lab\$ cat greeting.txt Hello World This is a test Goodbye World</pre>                                                                         |
| 9.  | Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines. |                                                                                                                                                                       |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                         |
|     | nano greeting.txt                                                                               |                                                                                    |
| 10. | Display the modified contents of "greeting.txt" using cat.                                      |                                                                                                                                                                       |
|     | <b>Command</b>                                                                                  | <b>Output</b>                                                                                                                                                         |
|     | cat greeting.txt                                                                                | <pre>rick@secr2043:~/os_lab\$ nano greeting.txt rick@secr2043:~/os_lab\$ cat greeting.txt Hello Yu Hin This is a test Goodbye Yu Hin rick@secr2043:~/os_lab\$  </pre> |
|     | <b>Total Mark:</b>                                                                              |                                                                                                                                                                       |

### ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

|    |                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Command                                                                                                                                                                                                                                                                                                                                                          | Output                                                                                                                                                                                                                                                                                                                                                                                                         |
|    | <pre>nano hello.c #include &lt;stdio.h&gt; int main() {     printf("Hello, World!\n");     return 0; }</pre>                                                                                                                                                                                                                                                     |  The screenshot shows a terminal window with the command 'nano hello.c' entered. Below it, the nano editor interface is visible, showing the code for 'hello.c' which includes <stdio.h> and has a main function that prints 'Hello, World!\n' and returns 0. The file name 'hello.c' is shown at the bottom of the editor. |
| 2. | Compile the program "hello.c" using gcc and generate an executable file named "hello".                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Command                                                                                                                                                                                                                                                                                                                                                          | Output                                                                                                                                                                                                                                                                                                                                                                                                         |
|    | <pre>gcc hello.c -o hello</pre>                                                                                                                                                                                                                                                                                                                                  | <pre>rick@secr2043:~/os_lab\$ gcc hello.c -o hello</pre>                                                                                                                                                                                                                                                                                                                                                       |
| 3. | Run the executable file "hello" and display its output.                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Command                                                                                                                                                                                                                                                                                                                                                          | Output                                                                                                                                                                                                                                                                                                                                                                                                         |
|    | <pre>./hello</pre>                                                                                                                                                                                                                                                                                                                                               | <pre>rick@secr2043:~/os_lab\$ ./hello Hello, World! rick@secr2043:~/os_lab\$</pre>                                                                                                                                                                                                                                                                                                                             |
| 4. | Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Command                                                                                                                                                                                                                                                                                                                                                          | Output                                                                                                                                                                                                                                                                                                                                                                                                         |
|    | <pre>nano #include &lt;stdio.h&gt; #include &lt;stdlib.h&gt;  int main(int argc, char *argv[]) {     if (argc != 3) {         printf("Usage: %s &lt;number1&gt; &lt;number2&gt;\n", argv[0]);         return 1;     }     int num1 = atoi(argv[1]);     int num2 = atoi(argv[2]);     int sum = num1 + num2;     printf("Sum: %d\n", sum);     return 0; }</pre> | <pre>rick@secr2043:~/os_lab\$ nano sum.c</pre>                                                                                                                                                                                                                                                                                                                                                                 |

|    |                                                                                                                                                                                    |                                                                                           |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|    | <pre>     }      int num1 = atoi(argv[1]);     int num2 = atoi(argv[2]);      int result = num1 + num2;      printf("%d + %d = %d\n", num1, num2, result);      return 0; } </pre> |         |
| 5. | Compile the program "sum.c" using gcc and generate an executable file named "sum".                                                                                                 |                                                                                           |
|    | <b>Command</b>                                                                                                                                                                     | <b>Output</b>                                                                             |
|    | gcc sum.c -o sum                                                                                                                                                                   | <pre> rick@secr2043:~/os_lab\$ gcc sum.c -o sum </pre>                                    |
| 6. | Run the executable file "sum" with 10 and 20 as arguments and display its output.                                                                                                  |                                                                                           |
|    | <b>Command</b>                                                                                                                                                                     | <b>Output</b>                                                                             |
|    | ./sum 10 20                                                                                                                                                                        | <pre> rick@secr2043:~/os_lab\$ ./sum 10 20 10 + 20 = 30 rick@secr2043:~/os_lab\$   </pre> |
| 7. | Run the executable file "sum" with -5 and 15 as arguments and display its output.                                                                                                  |                                                                                           |
|    | <b>Command</b>                                                                                                                                                                     | <b>Output</b>                                                                             |
|    | ./sum -5 15                                                                                                                                                                        | <pre> rick@secr2043:~/os_lab\$ ./sum 5 15 5 + 15 = 20 rick@secr2043:~/os_lab\$   </pre>   |
| 8. | Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".                                       |                                                                                           |
|    | <b>Command</b>                                                                                                                                                                     | <b>Output</b>                                                                             |
|    | <pre> nano echo.c #include &lt;stdio.h&gt; #include &lt;string.h&gt;  int main() { </pre>                                                                                          | <pre> rick@secr2043:~/os_lab\$ nano echo.c  </pre>                                        |

|     |                                                                                                                                                               |                                                                                                                        |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
|     | <pre> char name[50];  printf("Enter text: "); fgets(name, 50, stdin);  name[strcspn(name, "\n")] = '\0';  printf("Text input: %s", name);  return 0; } </pre> |                                      |
| 9.  | Compile the program "echo.c" using gcc and generate an executable file named "echo".                                                                          |                                                                                                                        |
|     | <b>Command</b>                                                                                                                                                | <b>Output</b>                                                                                                          |
|     | gcc echo.c -o echo                                                                                                                                            | <pre> rick@secr2043:~/os_lab\$ gcc echo.c -o echo </pre>                                                               |
| 10. | Run the executable file "echo", type "Hello World" as input and display its output.                                                                           |                                                                                                                        |
|     | <b>Command</b>                                                                                                                                                | <b>Output</b>                                                                                                          |
|     | ./echo<br>Hello World                                                                                                                                         | <pre> rick@secr2043:~/os_lab\$ ./echo Enter text: Hello World Text input: Hello Worldrick@secr2043:~/os_lab\$   </pre> |
|     | <b>Total Mark:</b>                                                                                                                                            |                                                                                                                        |